



Team Details

Team Name:

TEAM NAME REQUIRED

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	Team Leader	Vishnu	YEAR REQUIRED
2	Member 1	Not supplied	-
3	Member 2	Not supplied	-
4	Member 3	Not supplied	-

i A team can have up to 4 members including the team leader. Add rows if necessary.

COLLEGE NAME

COLLEGE REQUIRED BEFORE
SUBMISSION

TEAM LEADER CONTACT NUMBER

PHONE REQUIRED

TEAM LEADER EMAIL ADDRESS

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KLA PS01: Restore Degraded Inspection Images



Joint denoising and 2x super-resolution for grayscale NPY inspection data

INPUT, TARGET AND RISK

Map noisy 128x128 arrays to clean 256x256 arrays. The release has 3,200 paired training samples and 400 test inputs. The key risk is restoring fine evidence without inventing inspection structures.



DAF-Restormer Restores Structure, Not Texture



An accuracy-first transformer infers the degradation, restores spatial and frequency evidence, and avoids adversarial texture synthesis.

DEGRADATION-AWARE RESTORATION

Condition transformer blocks on a learned degradation prompt, then refine spatial and frequency features jointly.

ACCURACY MODE

Predict clean LR structure and uncertainty, upscale by 2x, then average eight dihedral views for robust output.



One Reproducible Command Runs the Full Pipeline



NPY input -> DAF-Restormer -> 8-view geometric ensemble -> clamp $[0,1]$ -> same-name float32 256x256 NPY output

TRAINING AND INFERENCE

Train on a fixed 2,880/320 split with seed 42 and fidelity-first losses; refine perceptually at low learning rate. infer.py auto-loads the selected checkpoint, batches GPU inference, and preserves filenames.



Innovation with a Verified Accuracy Win



The model estimates what degraded each image, restores in two domains, and exposes uncertainty.

MODEL DESIGN

Global degradation prompt + spatial noise map + frequency-gated bottleneck + auxiliary clean-LR and uncertainty heads.

MEASURED ADVANTAGE

28.4713 dB PSNR | 0.771162 SSIM | 0.228489 LPIPS. Beats the converged Restormer control on all three quality metrics.



Restoration Value and Readiness



Improve visible fidelity and 2x resolution while preserving a no-GAN safety posture for inspection evidence.

PRIMARY IMPACT

Cleaner edges support review and downstream defect analysis; adversarial hallucination is deliberately avoided.

VALIDATED OUTPUT

400/400 files passed filename, 256x256 shape, float32 dtype, finite-value, and [0,1] range checks.



Technology, Feasibility and H100 Evaluation



A portable PyTorch evaluator needs only input and output directories; the same path runs on CPU, T4 or H100.

REPRODUCIBLE STACK

Python 3.12, PyTorch 2.6, Git LFS checkpoint, and a pinned transitive lock. train.py reproduces training; infer.py auto-loads weights. The T4-validated CUDA path transfers unchanged to H100.



PyTorch + DAF model



T4 trained; H100-ready



Git LFS + pytest



Code and Executable Evidence



GITHUB REPOSITORY

github.com/IAM-MUKUND/image-restoration-project



RESTORED-OUTPUT EVIDENCE

[artifacts/remote-runs/daf-final-submission/](#) (400 outputs + validation)



Research & References



RESEARCH FOUNDATION

Fidelity-first transformer restoration with explicit degradation conditioning.

DAF-Restormer combines Restormer attention/GDFN blocks with learned degradation prompts, spatial noise estimation, FFT-domain gating, auxiliary supervision, and dihedral self-ensemble.



REFERENCES

Core architectures and official challenge source.

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arXiv:2108.10257

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